Elementary Problems
Existence Problem

Version 1.0

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Basic Work

1. (IMO shortlist 1971) Let $n \geqslant 2$ be a natural number. Find a way to assign natural numbers to the vertices of a regular $2n$-gon such that the following conditions are satisfied:

   (1) only digits 1 and 2 are used;

   (2) each number consists of exactly n digits;

   (3) different numbers are assigned to different vertices;

   (4) the numbers assigned to two neighbouring vertices differ at exactly one digit.

2. (Baltic Way 2001) Let $n \geqslant 2$ be a positive integer. Find whether there exist n pairwise nonintersecting nonempty subsets of $\{1, 2, 3, \ldots\}$ such that each positive integer can be expressed in a unique way as a sum of at most n integers, all from different subsets.

3. (Baltic Way 1995) A polygon with $2n+1$ vertices is given. Show that it is possible to assign numbers $1, 2, \ldots, 4n+2$ to the vertices and midpoints of the sides of the polygon so that for each side the sum of the three numbers assigned to it is the same.

4. Let n be a positive integer. There are $2n+1$ zeros and $2n+1$ ones lying in a circle in some order. Prove that there exists a number whose neighbours are both zeros.

5. Each positive integer is painted with black or white. Prove that for any positive integer N, there exist positive integers $m, n > N$ such that the integers $m, n, m+n$ are of the same colour.

6. (Russian MO 2007 Grade 9 Problem 5) Two numbers are written on each vertex of a convex 100-gon. Prove that it is possible to remove a number from each vertex so that the remaining numbers on any two adjacent vertices are different.

   (Remark: We may assume the two numbers written on each vertex are different.)

7. (Finnish National High School Mathematics Competition 2004 Problem 4) The numbers $2005!+2$, $2005!+3$, $\ldots$, $2005!+2005$ form a sequence of 2004 consecutive integers, none of which is a prime number. Does there exist a sequence of 2004 consecutive integers containing exactly 12 prime numbers?

8. (IMO shortlist 1988) Around a circular table an even number of persons have a discussion. After a break they sit again around the circular table in a different order. Prove that there are at least two people such that the number of participants sitting between them before and after a break is the same.

   (Remark: If $A, B, C, D, E, \ldots$ sit in this order clockwise initially, and they sit in the order $C, B, A, D, E, \ldots$ after the break, then A and C are not considered as a pair of persons satisfying the condition.)

9. (Russian MO 1995 Grade 9 Problem 4) Can the numbers from 1 to 81 be written in a 9×9 board, so that the sum of numbers in each 3×3 square is the same?

Pigeonhole Principle

10. Let n be a positive integer. Prove that if S is a subset of $\{1, 2, \ldots, 2n\}$ consisting of $n+1$ elements, then S contains two elements which are relatively prime.

11. Suppose there are 17 rooks on an 8×8 chessboard. Prove that there are 3 rooks such that no 2 of them threaten each other. The rooks threaten each other if they are put on two cells in the same row or the same column. There could be some other rooks in between.

12. Suppose there are 20 pairwise distinct positive integers not exceeding 70. Prove that there are 4 distinct pairs of these positive integers having the same difference.

    (Remark: Two of the pairs may share a common positive integer but not both.)

13. A chessmaster has 77 days to prepare for a tournament. He wants to play at least one game per day, but not more than 132 games in total. Prove that there is a sequence of successive days on which he plays exactly 21 games.

14. (Erdős-Szekeres theorem) Let m and n be positive integers. Prove that any sequence of at least $(m-1)(n-1)+1$ terms of real numbers contains a monotonic increasing subsequence of length m or a monotonic decreasing subsequence of length n.

15. (IMO shortlist 1966) Given a finite sequence of integers $a_1, a_2, \ldots, a_n$ for $n \geqslant 2$, show that there exists a subsequence $a_{k_1}, a_{k_2}, \ldots, a_{k_m}$, where $1 \leqslant k_1 < k_2 < \cdots < k_m \leqslant n$, such that the number $a_{k_1}^2 + a_{k_2}^2 + \cdots + a_{k_m}^2$ is divisible by n.

16. (IMO shortlist 1971) Natural numbers from 1 to 99 (not necessarily distinct) are written on 99 cards. It is given that the sum of the numbers on any subset of cards (including the set of all cards) is not divisible by 100. Show that all the cards contain the same number.

17. (IMO shortlist 1977) Let B be a set of k sequences each having n terms equal to 1 or -1. The product of two such sequences $(a_1, a_2, \ldots, a_n)$ and $(b_1, b_2, \ldots, b_n)$ is defined as $(a_1b_1, a_2b_2, \ldots, a_nb_n)$. Prove that there exists a sequence $(c_1, c_2, \ldots, c_n)$ such that the intersection of B and the set containing all sequences from B multiplied by $(c_1, c_2, \ldots, c_n)$ contains at most $\dfrac{k^2}{2^n}$ sequences.

18. (IMO shortlist 1972) Show that for any $n \not\equiv 0 \pmod{10}$ there exists a multiple of n not containing the digit 0 in its decimal expansion.

19. (IMO 1972 Problem 1) Prove that from a set of ten distinct two-digit numbers, it is possible to select two disjoint subsets whose members have the same sum.

20. (Poland MO Finals 2001 Problem 6)

    Given positive integers $n_1 < n_2 < \cdots < n_{2000} < 10^{100}$, prove that we can choose from the set $\{n_1, \ldots, n_{2000}\}$ nonempty, disjoint sets A and B which have the same number of elements, the same sum and the same sum of squares.

21. (India MO 2006 Problem 4) Some 46 squares are randomly chosen from a 9×9 chessboard and coloured in red. Show that there exists a 2×2 block of 4 squares of which at least three are coloured in red.

22. (Belarus MO 2007 Problem 4) Each point of a circle is painted in one of the N colours ($N \geqslant 2$). Prove that there exists an inscribed trapezoid such that all its vertices are painted the same colour.

23. Let $S = \{(x, y) : x, y = 0, 1, 2, \ldots, 7\}$ be a set of lattice points in the coordinate plane. We arbitrarily choose 196 distinct rectangles each of which has all 4 vertices in S and has sides parallel to the coordinate axes. Prove that there is a chosen rectangle lying inside another chosen rectangle.

24. The integers from 1 to 17 are put in the cells of a 17×17 table such that each cell contains one integer and each integer appears exactly 17 times. Prove that there exists a row or a column consisting of at least 5 different integers.

25. (USAMO 1986 Problem 2) During a certain lecture, each of five mathematicians fell asleep exactly twice. For each pair of mathematicians, there was some moment when both were asleep simultaneously. Prove that, at some moment, three of them were sleeping simultaneously.

26. (IMO 1978 Problem 6) An international society has its members from six different countries. The list of members contains 1978 names, numbered $1, 2, \ldots, 1978$. Prove that there is at least one member whose number is the sum of the numbers of two members from his own country, or twice as large as the number of one member from his own country.

Proof by Contradiction

27. Let n be a positive integer. There are n red cards and n blue cards. A positive integer is written on each card. Suppose the sum of integers on the cards with the same colour does not exceed n^2. Prove that there are 2 cards having the same integer written on them.

28. Let n be an odd positive integer greater than 1. We colour the vertices of a regular polygon with n sides with red, blue and green such that every 2 consecutive vertices have distinct colours. Prove that there exist 3 consecutive vertices whose colours are pairwise distinct.

29. Find all nonempty finite sets S of real numbers such that for any $a \in S$, we have $a + |a| \in S$.

30. There are some people arranged in a circle. Each person has a hat in one of the three colours, red, blue or green. It is known that there are 20 people with red hats each of whom is a neighbour of a person with a green hat, and there are 25 people with blue hats each of whom is a neighbour of a person with a green hat. Prove that there exists a person whose two neighbours both have green hats.

31. A 6×6 chessboard is tiled completely by some dominoes (1×2 or 2×1). Prove that we can divide the chessboard into two smaller rectangular chessboards without cutting any domino.

32. Let n be a positive integer. There are $3n$ points on the circle which divide the circle into n arcs of length 1, n arcs of length 2 and n arcs of length 3. Prove that there are 2 points which are the endpoints of a diameter of the circle.

33. Let n be a positive integer. There is a finite number of coins such that the denomination of each coin can only be $\dfrac{1}{k}$ for some integer $k \geqslant 2$. If the total value of the coins is less than $\dfrac{n+1}{2}$, prove that we can partition the coins into at most n groups so that the total value of the coins in each group is less than 1.

34. (Iran MO 2nd Round 2018 Problem 5) Lamps of the hall switch by only five keys. Every key is connected to one or more lamp(s). By switching every key, all connected lamps will be switched too. We know that no two keys have same set of connected lamps with each other. At first all of the lamps are off. Prove that someone can switch just three keys to turn at least two lamps on.

    (Remark: Here, just three keys means exactly three keys.)

35. (IMO 1997 Problem 4)

    An $n \times n$ matrix whose entries come from the set $S = \{1, 2, \ldots, 2n-1\}$ is called a silver matrix if, for each $i = 1, 2, \ldots, n$, the ith row and the ith column together contain all elements of S. Show that

    (a) there is no silver matrix for $n = 1997$;

    (b) silver matrices exist for infinitely many values of n.

Record

v1.0: problems 1–35